

Statistics for Data Science

Understanding and quantifying uncertainty



From Statistics to Probability

Last class: we used a sample to learn about a population.

A survey of 500 students found an average of 6.7 hours of sleep.

6.5

Sample 1

6.7

Observed sample

6.9

Sample 2

Why might another sample of 500 give a different mean?

Because data contain randomness and uncertainty.

Why Do We Need Probability?

Data science rarely gives us complete certainty.

- Will a customer click an advertisement?
- Is an email spam?
- Will a student pass a course?
- Will a customer cancel a subscription?
- Is a transaction fraudulent?
- What will tomorrow's demand be?

Instead of “This will definitely happen,” we can say:

“There is an 85% probability this will happen.”

How Is Probability Used in Data Science?

INFERENCE

How well does a sample represent a population?

CLASSIFICATION

$P(\text{Spam} \mid \text{Email Features})$

A/B TESTING

Is a measured difference real or random variation?

PREDICTION

What is the probability a customer will churn?

DECISIONS

Should a bank approve a loan at this risk?

MACHINE LEARNING

Many models report probabilities, not certainties.

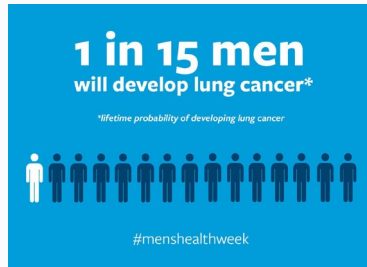
Probability turns uncertainty into information we can use.

- Estimate population parameters
- Build confidence intervals
- Perform hypothesis testing
- Understand sampling distributions

How Is Probability Used in Data Science?

HOSPITAL DATA

Patient_ID	Age	Gender	Blood_Type	Chronic_Condition	BMI	Annual_Visits	Avg_Duration	Total_Spending
P10001	45.00	Male	A+	0	20.00	3	1.960	5.850
P10001	60.00	Female	A+	1	29.56	1	7.920	6.880
P10002	30.00	Male	O+	1	36.00	4	7.020	8.540
P10003	30.00	Female	A-	0	31.20	1	6.210	Public
P10004	45.00	Female	AB+	0	35.65	6	8.810	7.890
P10005	30.00	Male	A+	1	55.65	1	6.340	Private
P10006	45.00	Female	AB-	0	32.08	3	5.840	Poor
P10007	50.00	Male	AB	0	48.08	5	5.280	Pmate
P10008	60.00	Female	A-	1	52.65	5	7.660	Public
P10009	45.00	Male	A+	0	55.64	1	9.800	None
P10100	45.00	Female	AB	0	52.08	3	6.370	Public
P10101	60.00	Male	AB	0	53.50	5	8.520	Base
P10102	45.00	Female	AB	4	55.50	1	7.070	Pratic
P10103	45.00	Male	Am	0	55.50	2	3.050	Public

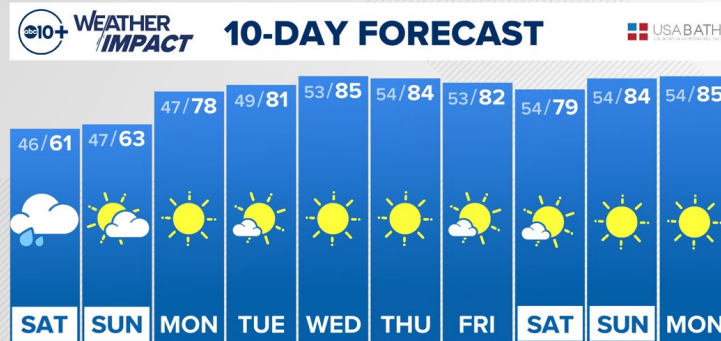


Household Income	Gambling Post-Year n=3634		Panel Type		Online Only n=1107		Phone n=1527		Online Only n=134 (5.3)		Venue Type Land-Based Only n=1615 (75.4)		Online & Land-Based n=477 (19.2)	
	%Yes	N	%Yes	N	%Yes	N	%Yes	N	%Yes	N	%Yes	N	%Yes	N
<= \$15,000*	58.0*	191	61.4	132	52.5	59	7.1	8	77.7	87	15.2	8		
\$15,000 - \$29,999	71.0	310	74.9*	207	63.7	102	3.2	7	81.8*	180	15.0	33		
\$30,000 - \$49,999	73.0	460	80.2*	329	55.0	131	6.9	23	74.3	249	18.8	63		
\$50,000 - \$69,999	76.1	548	83.9*	355	62.1	195	5.0	21	72.6	304	22.4	94		
\$70,000 - \$99,999	74.9	626	80.7*	415	63.3	211	5.3	25	74.0	347	20.7	97		
\$100,000 - \$124,999	75.8	400	82.6*	259	63.1	140	5.0	15	69.0	209	26.1*	79		
\$125,000 - \$149,999	73.3	221	83.3*	120	61.4	101	6.8	11	67.9	130	25.3	41		
\$150,000 or more	68.2	412	75.2	164	66.6	247	6.0	17	78.5	223	15.5	44		
Prefer not to answer	50.5	467	61.6	126	45.2	341	3.4	8	88.8	206	7.8	18		

*p < .02

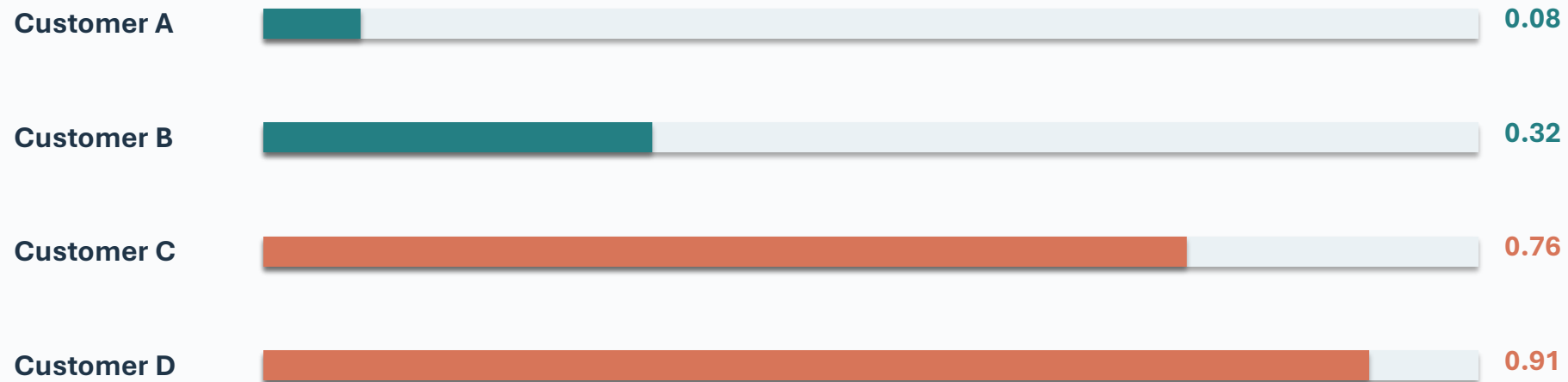


Gemeenten	Datum Tijd	Datum	Tijd	Temp	Gevoels Temp	Kans op Zon	Kans op Buien	Neerslag in mm	Wind Richting	Wind Kracht
Almere	27-6-2014 00:00	27-6-2014	00:00	12	12	0	0	0	N	2
Almere	27-6-2014 01:00	27-6-2014	01:00	11	11	0	2	0	N	2
Almere	27-6-2014 02:00	27-6-2014	02:00	11	11	0	0	0	NO	2
Almere	27-6-2014 03:00	27-6-2014	03:00	12	12	0	0	0	NO	1
Almere	27-6-2014 04:00	27-6-2014	04:00	11	11	0	0	0	O	2
Almere	27-6-2014 05:00	27-6-2014	05:00	12	12	0	0	0	O	2
Almere	27-6-2014 06:00	27-6-2014	06:00	14	14	0	0	0	ZO	1
Almere	27-6-2014 07:00	27-6-2014	07:00	15	15	56	0	0	ZW	1
Almere	27-6-2014 08:00	27-6-2014	08:00	16	16	62	0	0	ZW	2
Almere	27-6-2014 09:00	27-6-2014	09:00	17	17	62	5	0	W	1
Almere	27-6-2014 10:00	27-6-2014	10:00	17	17	27	10	0	W	1
Almere	27-6-2014 11:00	27-6-2014	11:00	19	19	20	27	0	2 W	2
Almere	27-6-2014 12:00	27-6-2014	12:00	19	19	15	42	0	4 ZW	2
Almere	27-6-2014 13:00	27-6-2014	13:00	19	19	0	58	0	4 ZW	2
Almere	27-6-2014 14:00	27-6-2014	14:00	19	19	0	66	0	5 Z	2
Almere	27-6-2014 15:00	27-6-2014	15:00	20	20	0	65	0	5 ZW	2
Almere	27-6-2014 16:00	27-6-2014	16:00	20	20	0	59	0	5 ZW	2
Almere	27-6-2014 17:00	27-6-2014	17:00	20	20	0	53	0	3 ZW	2
Almere	27-6-2014 18:00	27-6-2014	18:00	19	19	0	55	0	2 ZW	2
Almere	27-6-2014 19:00	27-6-2014	19:00	19	19	0	47	0	1 W	2
Almere	27-6-2014 20:00	27-6-2014	20:00	19	19	0	42	0	W	2
Almere	27-6-2014 21:00	27-6-2014	21:00	18	18	0	39	0	W	2
Almere	27-6-2014 22:00	27-6-2014	22:00	16	16	43	33	0	W	2
Almere	27-6-2014 23:00	27-6-2014	23:00	15	15	0	22	0	W	2
Almere	28-6-2014 00:00	28-6-2014	00:00	15	15	0	16	0	ZW	2



A Data Science Example: Churn Risk

A model estimates the probability that each customer will cancel.



What does 0.76 mean?

It represents uncertainty. It does not guarantee that Customer C will leave.

Random Experiment

A random experiment produces an outcome that cannot be predicted with certainty.

01 Tossing a coin

02 Rolling a die

03 Selecting a customer

04 Observing whether a user clicks an ad

05 Pulling colored ball out of the bag



**We can know the
possible outcomes
without knowing which
one will occur.**

That uncertainty is what
probability describes.

Outcome and Sample Space

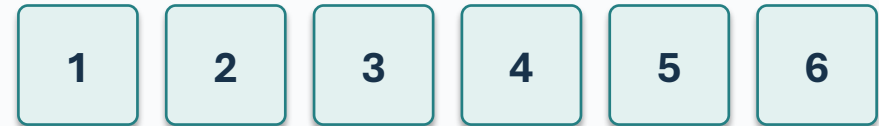
An outcome is one possible result of a random experiment.

Roll a six-sided die

One outcome: 3

Sample space S: the set of all possible outcomes

$S = \{1, 2, 3, 4, 5, 6\}$



Coin toss sample space

$S = \{H, T\}$

One toss gives either Heads or Tails.

What Is an Event?

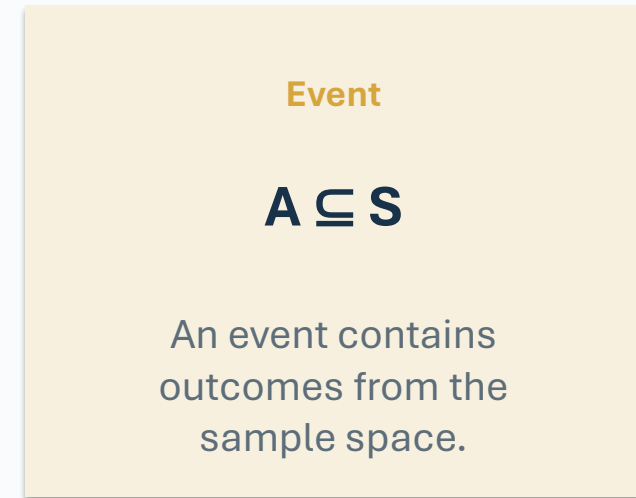
An event is a set of outcomes of an experiment to which a probability is assigned. In set notation, an event is a subset of S .

Die sample space



A: roll an even number = {2, 4, 6}

B: roll a number greater than 4 = {5, 6}



- E represents event
- $P(E)$ is the probability that the event E occur.
- A situation where E might happen (success) or might not happen (failure) is called a trial.

Events in Data Science

Events describe outcomes that matter in a real question.

A Customer makes a purchase

B Transaction is fraudulent

C Patient's test is positive

D Student scores above 80

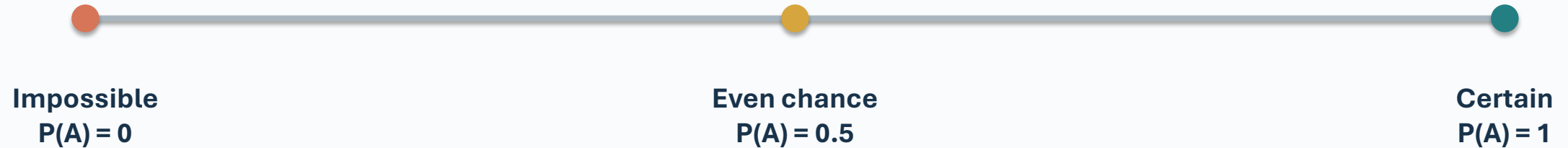
Probability lets us ask how likely each event is.

$P(A)$ $P(B)$ $P(C)$ $P(D)$

Probability

Probability is a numerical measure of how likely an event is to occur.

$$0 \leq P(A) \leq 1$$



More likely events have larger probabilities.

Two Ways to Understand Probability

Both approaches quantify uncertainty, but they start from different evidence.

THEORETICAL

If the probability of an event can be calculated even **before** the actual happening of the event, that is, even before conducting the experiment, it is called **Mathematical probability**

“If we keep using the examples of tossing of fair coin, dice etc., we can state the answer in advance (prior), without tossing of coins or without rolling the dice etc.,”

EMPIRICAL

If the probability of an event can be determined only **after** the actual happening of the event, it is called **Statistical probability(or a posteriori probability)**.

“If a coin is tossed 10 times we may get 6 heads and 4 tails or 4 heads and 6 tails or any other result. In these cases, the probability of getting a head is not 0.5 as we consider in Mathematical probability.”

Theoretical Probability

When outcomes are equally likely:

$$P(A) = \text{favorable outcomes} / \text{total possible outcomes}$$

Fair die

Event: roll an even number

Favorable outcomes: {2, 4, 6}

P(Even)

$$3 / 6 = 0.5$$

This result relies on the fair-die assumption.

Empirical Probability

Estimate probability from observations:

$$P(A) \approx \text{times A occurred} / \text{total observations}$$

Among 10,000 customers, 850 canceled.

Estimated churn probability

$$850 / 10,000 = 0.085 \text{ (8.5\%)}$$

Empirical probability is based on observed data.

Theoretical vs. Empirical Probability

Theoretical	Empirical
Based on a model	Based on observed data
No experiment required	Requires observations
Exact under assumptions	An estimate
Example: fair die	Example: customer churn

A fair coin has $P(\text{Heads}) = 0.50$.

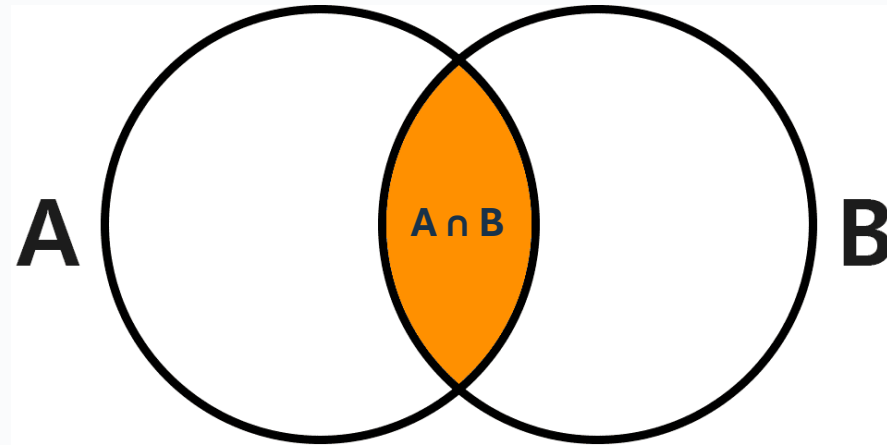
In 100 tosses, we might observe 56 heads: $\hat{p} = 0.56$.

The difference comes from random variation. More tosses tend to stabilize the observed proportion.

Intersection of Events

A: customer is under 30 B: customer purchases a product

Sample space S

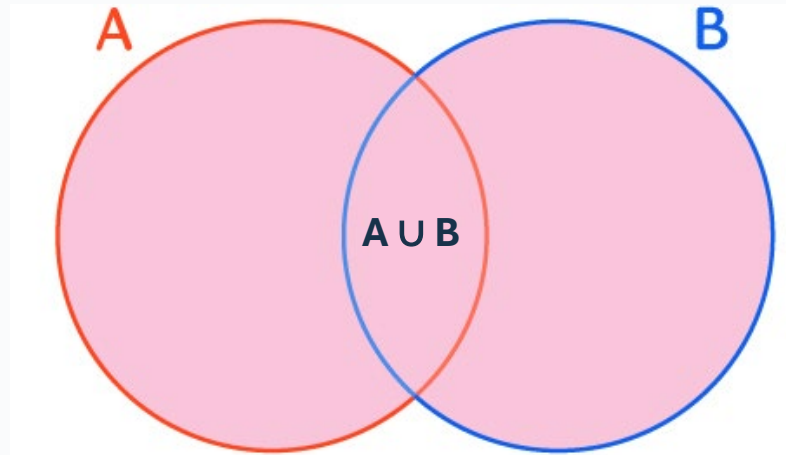


$A \cap B$ means the customer is under 30 AND makes a purchase.

Union of Events

A: purchases Product A B: purchases Product B

Sample space S



$A \cup B$ means the customer purchases A, B, or both.

Independent Events

Two events are independent when one does not change the probability of the other.

Flip a fair coin twice

HH

HT

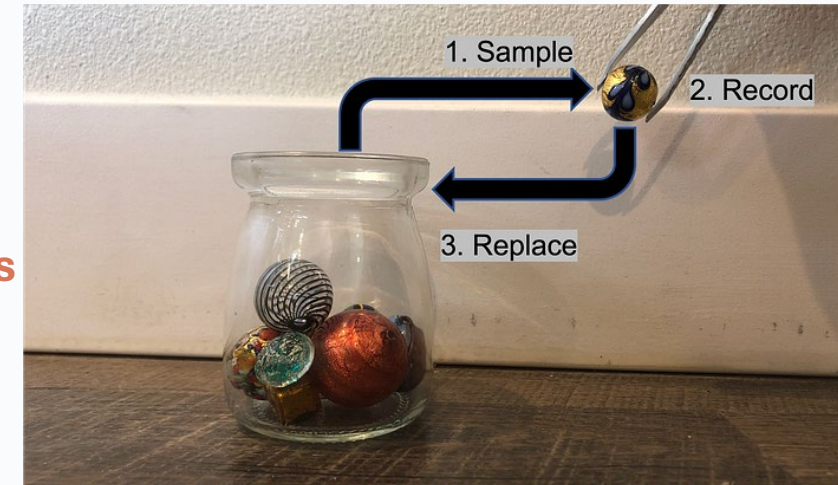
TH

TT

A: first toss is Heads

B: second toss is Heads

Knowing A occurred does not change $P(B) = 0.5$.



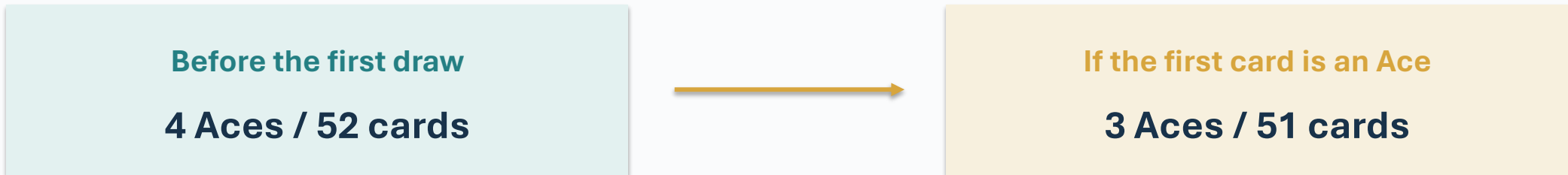
Each sampling is an independent event

$$P(A \cap B) = P(A)P(B) = 0.5 \times 0.5 = 0.25$$

Dependent Events

Two events are dependent when knowing one occurred changes the probability of the other.

Draw two cards without replacement.



A: first card is an Ace B: second card is an Ace

The first draw changes the outcomes left for the second draw, so A and B are dependent.

Dependent Events

Two events are dependent when knowing one occurred changes the probability of the other.



Each sampling is a dependent event

Data Science Example: Independence?

A: a customer is younger than 25.

B: a customer uses TikTok daily.

Are A and B necessarily independent?

Probably not. Knowing a person's age may change the probability that they use TikTok.

Independence is an assumption to justify, not a default.

Challenge: Independent or Dependent?

For each pair, decide whether the events are likely independent. Explain your reasoning.

1. A = It rains today

B = It rains tomorrow

2. A = First coin toss is Heads

B = Second coin toss is Heads

3. A = Customer purchased before

B = Customer responds to a promotion

4. A = Student attends class regularly

B = Student earns a high grade

Some answers depend on context. Focus on whether knowing one event changes the probability of the other.

A Different Question

Suppose $P(\text{Customer Purchases}) = 0.20$.

Now we learn that the customer has purchased from us before.

Before the new information

$P(\text{Purchase})$

Given previous purchase

$P(\text{Purchase} \mid \text{Previous})$



The condition changes the question we ask.

Conditional Probability

Probability of event A occurring, given that event B has already occurred:

$$P(A | B) = P(A \cap B) / P(B)$$

In the full sample space, $P(B) = 0.65$ and $P(A \cap B) = 0.30$.

$A \cap B = 0.30$

Rest of B

Outside B

$$P(A | B) = 0.30 / 0.65 = 0.46$$

Once B is known, B becomes the relevant sample space.

Conditional Probability and Independence

If A and B are independent, knowing B gives no new information about A.

$$P(A | B) = P(A)$$

If the probabilities differ, the events are dependent.

$P(A | B) \neq P(A)$ suggests A and B are dependent

Conditional probability gives another way to understand independence.

Why Bayes' Theorem?

These probabilities look similar, but they answer different questions.

$P(\text{Positive Test} \mid \text{Disease})$

$P(\text{Disease} \mid \text{Positive Test})$

The first asks how often a diseased patient tests positive.

The second asks how likely disease is after a positive result.

- If a medical test is positive, what is the probability the patient actually has the disease?
- If an email contains certain words, what is the probability it is spam?
- If a customer clicked an ad, what is the probability they are interested in the product?
- These questions have the structure:

$$P(A|B)$$

Bayes' Theorem

Bayes' theorem updates a probability when new evidence becomes available.

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Diagram illustrating Bayes' Theorem with labels for each term:

- $P(A|B)$: Probability of A occurring given evidence B has already occurred
- $P(B|A)$: Probability of B occurring given evidence A has already occurred
- $P(A)$: Probability of A occurring
- $P(B)$: Probability of B occurring

PRIOR

 $P(A)$ 

EVIDENCE

Observe B



UPDATE

Use Bayes' theorem



POSTERIOR

 $P(A | B)$

Bayes' theorem is a rule for updating probabilities as evidence arrives.

Bayes Example: Fraud Detection

PREVALENCE

1% of transactions are fraudulent.

DETECTION

The system flags 90% of fraudulent transactions.

FALSE ALARMS

The system flags 5% of legitimate transactions.

A transaction is flagged.

What is the probability that it is actually fraudulent?

Is it 90%? Pause and make an estimate before calculating.

Bayes Example: Think in Counts

Start with 10,000 transactions.

Transaction type	Total	Flagged	Not flagged
Fraudulent	100	90	10
Legitimate	9,900	495	9,405
Total	10,000	585	9,415

Fraud $10,000 \times 0.01 = 100$

Flagged fraud $100 \times 0.90 = 90$

$P(\text{Fraud} \mid \text{Flagged}) = 90 / 585 \approx 15.4\%$

Why Is the Bayes Result Surprising?

Fraud is rare, so the base rate starts very low.



The flag raises our belief that the transaction is fraudulent.

But a flagged transaction is still more likely legitimate than fraudulent.

Always consider the base rate. This idea also matters for classification, precision, and recall.

Bayes Example: Medical Test

Suppose:

1% of people have a disease

Test accuracy:

- 99% positive if the person has the disease
- 5% false positive if the person does not have the disease

We want:

$$P(\text{Disease} \mid \text{Positive})$$

What Is a Random Variable?

A random variable assigns a numerical value to the outcome of a random experiment.

Usually written X

Roll a die

$X = \text{number shown}$

Possible values

$X \in \{1, 2, 3, 4, 5, 6\}$

Before the roll, X is unknown. After the roll, X takes one specific value.

What Is a Random Variable?

A random variable assigns a numerical value to the outcome of a random experiment.

Example: $\Omega = 5$ coin tosses = $\{ \langle \text{HHHHH} \rangle, \langle \text{HHHHT} \rangle, \langle \text{HHHTH} \rangle, \langle \text{HHHTH} \rangle, \dots \}$

We may just care about how many tails? Thus,

$$X(\langle \text{HHHHH} \rangle) = 0$$

$$X(\langle \text{HHHTH} \rangle) = 1$$

$$X(\langle \text{TTTHT} \rangle) = 4$$

$$X(\langle \text{HTTTT} \rangle) = 4$$

X only has 6 possible values: 0, 1, 2, 3, 4, 5

What is the probability that we end up with $k = 4$ tails?

$$\mathbf{P}(X = k) := \mathbf{P}(\{ \omega : X(\omega) = k \}) \quad \text{where } \omega \in \Omega$$

Event : getting exactly 4 tails

Random variable: number of tails

X = the number of tails in 5 coin tosses

Possible values of X:

$$X \in \{0, 1, 2, 3, 4, 5\}$$

Random Variables in Data Science

A random variable can represent a quantity we do not know in advance.

01 Customer spending

02 Number of purchases

03 Exam score

04 Website response time

05 Daily temperature

06 Number of clicks

07 Model prediction error

Before observation, the value is uncertain. Afterward, we record a specific value.

Discrete Random Variables

A discrete random variable takes a countable set of possible values.

Purchases

$$X \in \{0, 1, 2, 3, \dots\}$$

Website clicks

$$X \in \{0, 1, 2, \dots\}$$

Defective products

$$X \in \{0, 1, 2, \dots\}$$

Discrete variables usually describe counts.

Continuous Random Variables

A continuous random variable can take any value within an interval.

HEIGHT

172.35 cm

TEMPERATURE

21.73 °C

RESPONSE TIME

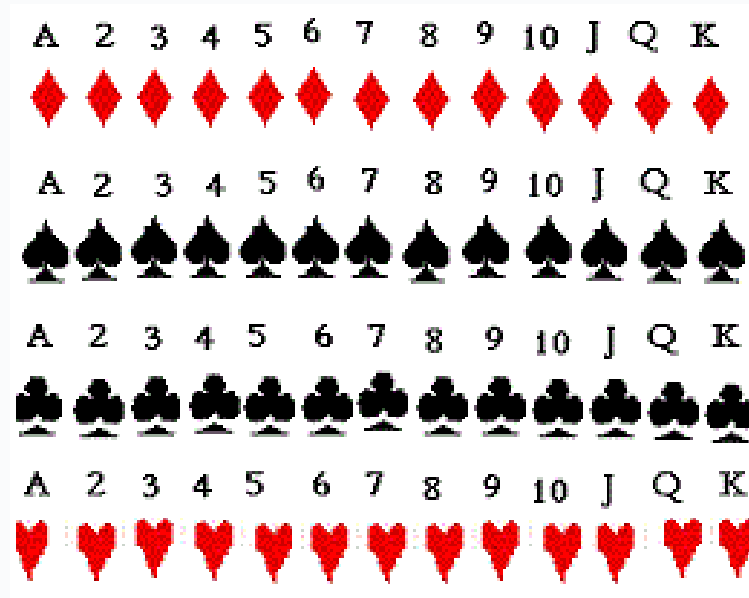
0.237 seconds

These values come from measurement and can include decimals.

Continuous data can be more precise as the measuring instrument improves.

Small Challenge

Two cards are drawn without replacement from a standard deck. What is the probability that: The first card is a diamond and the second card is the 3 of diamonds?



Connect Everything

A company studies churn among 2 million users and samples 5,000 customers.

Variable	Type
Age	Quantitative
Monthly spending	Quantitative
Number of logins	Quantitative
Subscription type	Qualitative
Churn (Yes/No)	Qualitative

- 1 What are the population and sample?
- 2 Which variables are quantitative and which are qualitative?
- 3 Let A = “customer churns.” What does $P(A)$ mean?
- 4 What could $P(A \mid \text{monthly spending} < \$10)$ mean?
- 5 Is churn a random outcome? How can probability support a decision?

Use evidence from the sample to think about customers beyond the sample.